

UE23CS320A – Capstone Project Approval

Project Title : AI-Based Dog Health Monitoring System Using
Dual-Input (Images + Symptoms) for Small, Medium, and
Large Breeds

Project ID :193

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Outline

Problem Statement

Problem statement

Pet owners often struggle to identify whether their pets are suffering from common or serious illnesses. Visiting a veterinarian for every minor concern can be costly and inconvenient. Many diseases show early signs in the form of symptoms (loss of appetite, vomiting, fatigue) and visible indicators (abnormal stool, urine color, skin conditions, or vomit appearance). There is currently no widely available system

that integrates both symptom descriptions and visual evidence to support early health assessments. We aim to develop a hybrid, multi-modal AI system that analyzes both symptom descriptions and pet images to predict possible diseases, highlight key evidence, and guide owners on whether they should seek veterinary care.

Scope and Feasibility study

This project aims to develop an AI-based dog health monitoring system using dual-inputs: images and symptoms. The system will support small, medium, and large dog breeds, identifying visible and behavioral health issues. Image analysis will detect physical signs like skin problems or injuries, while symptom input will capture behavioral changes. A fusion model will process both inputs to suggest

possible health conditions. The system will be accessible via a mobile or web interface. Technically feasible using existing AI models and datasets. Economically viable for a research or MVP phase. Operationally practical with clear user guidelines and disclaimers. Offers valuable assistance to pet owners and can reduce unnecessary vet visits.

Feasibility study

Technical Feasibility

- . AI models (CNN for images, NLP for symptoms) are already available and can be adapted.
- . Preprocessing pipelines and fusion models are **well-researched**.
- . Datasets (dog images + symptom reports) can be sourced or expanded.

2. Economic Feasibility

- . Development cost is manageable at **research / prototype level**.
- . Open-source AI tools reduce cost.
- . Commercial scalability possible through premium features.

3. Operational Feasibility

- . Simple mobile/web app ensures **easy adoption** by pet owners.
- . Clear guidelines and disclaimers prevent misuse.
- . Reduces unnecessary vet visits, saving **time and expenses**.

Applications/Use cases

- Early Disease Detection – Identifies common illnesses (e.g., skin infections, eye issues) in dogs through image and symptom analysis.
- Breed-Specific Health Monitoring – Offers tailored insights based on breed size (small, medium, large) and common health risks.
- Remote Health Checks – Assists pet owners in rural or remote areas where access to veterinary services is limited.
- Pet Care Guidance – Suggests home care tips or alerts when veterinary attention is necessary.

- Veterinary Pre-Screening Tool – Helps clinics triage nonemergency cases and prioritize urgent ones.
- Educational Tool for Pet Owners – Informs owners about symptoms and breed-specific health concerns.
- Telehealth Integration – Can be extended to support remote consultations with veterinarians.
- Pet Wellness Apps – Integrates into pet care apps to track ongoing health and alerts.
- AI Research in Veterinary Medicine – Contributes to the development of intelligent systems for animal healthcare.

Objectives:

- Develop an AI model capable of analyzing pet symptoms and images (vomit, stool, urine, skin, etc.)
- Fuse multi-modal inputs (symptoms + images) for improved prediction accuracy - Provide explainable AI outputs (highlight image regions or symptom factors influencing predictions)
- Design the system to be user-friendly for non-expert pet owners - Offer preventive health advice and alerts for serious issues.

Expected Deliverable

Deliverables:

- A dataset combining symptom descriptions and annotated pet images
- A trained hybrid AI model (CNN for images + ML classifier for symptoms)
- A prototype web/mobile application for pet owners
- Explainability module (Grad-CAM for images, SHAP for symptoms) - Final report and presentation
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Risks and Challenges :

- Limited availability of publicly accessible datasets combining symptoms and images
 - Variability in image quality due to different lighting and camera devices
 - Need for careful design to ensure predictions are advisory, not diagnostic
 - Balancing explainability with accuracy
 - Integration of multi-modal data sources

Capstone (Phase-I Phase-II, Phase-III) Project Timeline

- Capstone-I deliverables

Defined and refined problem statement

Literature survey on CNNs, symptom analysis, multimodal AI

Initial dataset collection (dog images, symptoms, breed data)

Dataset annotation and preprocessing plan

Interim report and presentation

- **Capstone-II deliverables**

System architecture and workflow diagram

Preprocessing pipelines for image and symptom data

Baseline AI models (CNN for images, NLP/rule-based for symptoms)

Prototype interface (image/symptom input + result display)

Progress report and demo presentation

- Capstone-III deliverables

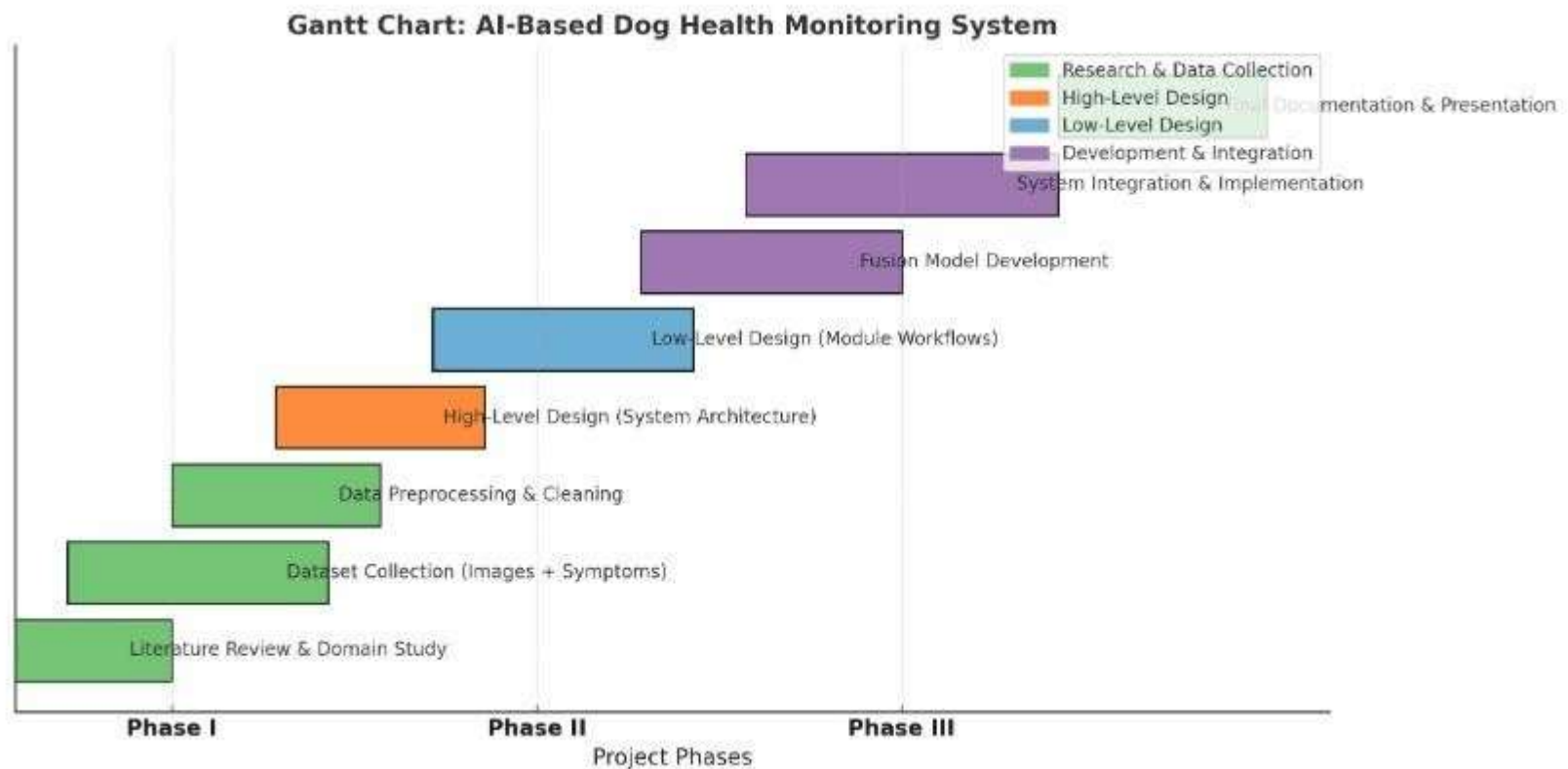
- Model optimization and evaluation results

- Functional recommendation/alert system

- Draft research paper (methodology + results)

- Usability testing report with user feedback

- Integrated dual-input AI model with breed-specific tuning



Gantt chart

Reference :

- Proceedings of the Third International Conference on Augmented Intelligence and Sustainable Systems (ICAISS-2025) IEEE Xplore Part Number: CFP25CB2-ART; ISBN: 979-8-3315-0724-4
- Proceedings of the International Conference on Intelligent Computing and Control Systems (ICICCS-2025) IEEE Xplore Part Number: CFP25K74-ART; ISBN: 979-8-3315-1208-8

Datasets:

Pet health link: <https://www.kaggle.com/datasets/sathwiknomula/animal-veterinary-health-dataset>

Symptoms link: <https://www.kaggle.com/datasets/yyzz1010/pet-health-symptoms-dataset>

Skin diseases link: <https://universe.roboflow.com/pethealth/dog-skin-disease-dataset-s8tt2>

Images link : <https://www.kaggle.com/datasets/amartya0roy/dogs-diseases>

Thank You